

Instructions – EAU1 Database

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+++Test Substance Chemical Structure+++:

Find the SMILES notation of the structure of the chemical test item. Look for: a structural diagram (e.g., a 2D depiction showing bonds and atoms), a chemical formula (e.g., $C_8H_{10}O_2$), a SMILES string or InChI code, or a CAS Registry Number (to be paired with known structure via external resources).

Common places where structure information may appear in a test report are Test Item Identity or Substance Identity section, Materials and Methods or formulation tables, and

Attachments/Appendices (e.g., Certificates of Analysis, Safety Data Sheets, Structural Appendices). Note: This may not be possible for UVCBs or when the test report only identifies the test item using a brand name or internal company code with no other explicit chemical identifiers.

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+++Test Substance Analogue+++:

When you are being asked to determine whether or not the test item in the toxicity study is the notified substance or an analogue substance, you need to determine if the test item identifiers match the notified substance identifiers (e.g., notified substance chemical name, notified substance CAS registry number, or optionally: SMILES, InChI, synonyms, other identifiers if available).

If the test item's chemical name or CAS registry number exactly matches the notified substance's name or CAS number, then the test item is the notified substance.

If neither the chemical name nor the CAS registry number of the test item matches the notified substance's identifiers, then the test item is an analogue. In this case, return the letter "A" in the field indicating analogue status. Also report the test item's CAS registry number, and report the SMILES notation of the test item's structure (if available in the PDF). If the PDF does not contain explicit SMILES, leave blank.

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+++Report format: Full Study (detailed report) or Summary+++:

The PDF document is either a full toxicity report or a summary/abstract/overview document. The defining features for each are provided below. Use the criteria to classify the document into one of two categories: “full toxicity report” or “summary document”.

For “full toxicity report”:

- Contains names of study personnel and signatures
- Indicates the test guideline followed and/or whether the test followed principles of Good Laboratory Practices
- Contains detailed, explicit sections describing information on most or all of the topics below for a single toxicological endpoint (e.g., repeated dose toxicity, genotoxicity, or skin sensitisation):
 - Test chemical:
 - physical nature, purity and physicochemical properties;
 - identification data.
 - Vehicle (if appropriate):
 - justification for choice of vehicle, if other than water.
 - Test animals or test system:
 - species/strain used;
 - number, age and sex of animals;
 - source, housing conditions, diet, etc.;
 - individual weights of animals at the start of the test.
 - justification for species
 - Test conditions:
 - rationale for dose level selection;
 - details of test chemical formulation/diet preparation, achieved concentration, stability and homogeneity of the preparation;
 - details of the administration of the test chemical;
 - conversion from diet/drinking water test chemical concentration (ppm) to the actual dose (mg/kg body weight/day), if applicable;
 - details of food and water quality; report whether (yes/no) blinding was applied in the study.
 - Optional endpoints investigated
 - list of optional endpoints investigated - samples cryopreserved (if applicable)
 - Results:
 - body weight/body weight changes;
 - food consumption, and water consumption, if applicable;
 - toxic response data by sex and dose level, including signs of toxicity; nature, severity and duration of clinical observations (whether reversible or not);
 - sensory activity, grip strength and motor activity assessments;
 - haematological tests with relevant base-line values;
 - clinical biochemistry tests with relevant base-line values;
 - body weight at euthanasia and organ weight data;
 - necropsy findings;
 - a detailed description of all histopathological findings;
 - absorption data if available;
 - statistical treatment of results, where appropriate.
 - Discussion of results
 - Conclusions
 - Multiple appendices with raw data or pathology worksheets

For “summary document”:

- Presents a high-level overview or abstract without detailed methods or raw data, or
- Contains information on many different toxicological endpoints, the details of which are minimal and span only a paragraph or two, or
- Focuses on conclusions, executive summaries, or key findings without full data sets (e.g., a NOAEL, EC3, “positive” or “negative” with minimal details on the methodology followed), or
- A document labelled “study summary”, “toxicological summary”, “brief report”, or
- There are no study personnel names or signatures

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+++Strain+++:

Identify the species and strain (for in vivo animal studies) *or* the biological test system used (for in vitro tests).

If this is an in vivo animal test:

- Find where the report states the species (e.g., rat, rabbit, mouse, guinea pig, dog, etc.) and the strain or stock of that species. Extract and return the strain.
- Common strains may include but not be limited to:
 - Sprague Dawley (e.g., Hsd:SD) - rat
 - Wistar / Wistar Hannover (e.g., RccHan:WIST / HsdHan:WIST) - rat
 - Long Evans (also called Hooded rat, e.g., Blu:(LE)) – rat
 - Fischer 344 (F344 / F344/N) – rat
 - Lewis (e.g., LEW/SnNH) – rat
 - New Zealand White (e.g., CrI:KBL(NZW)) - rabbit
 - Dunkin Hartley (e.g., HsdDhl:DH) - guinea pig
 - Hartley - guinea pig
 - CD-1 (CrI:CD-1) - mouse
 - ICR (e.g., Hsd:ICR, ICR/Jcl) - mouse
 - BALB/c - mouse
 - C57BL/6J - mouse
 - C57BL/6N - mouse

If this is an in vitro test that does not use whole animals:

- Determine whether the test uses bacterial strains (e.g., *S. typhimurium* or *E. coli* strains like TA98, TA1535, WP2uvra, etc.) or mammalian cell types/lines (e.g., ovary CHO cells, lung CHL cells, peripheral blood lymphocytes, Hep2 cells, mouse lymphoma L5178Y cells, BALB/c cells)
- Extract and return the strain of bacteria (for bacteria tests) or the type of mammalian cell used (for mammalian cell tests).

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+++Exposure type+++:

Determine the route of administration used for the in vivo animal test. Choose one of the following primary routes: oral, dermal, or inhalation. If the test is in vitro, leave blank.

If the route is oral, determine whether the substance was delivered by gavage (e.g., test item administered by intubation or stomach tube, typically with doses described in mL/kg or mg/kg) *or* by diet (e.g., mixed into feed, expressed as a % of diet or ppm) *or* via drinking water (e.g., dissolved/suspended in water, concentration given in mg/L or ppm). Report one of: “oral - gavage”, “oral - diet”, or “oral - drinking water”.

If the route is dermal, determine whether the application was occlusive (covered with impermeable barrier), semi-occlusive (covered with permeable dressing), or non-occluded (left open). Report one of: “dermal - occlusive”, “dermal - semi-occlusive”, or “dermal - non-occluded”.

If the route is inhalation, determine whether the exposure was **whole body** (animals exposed in an inhalation chamber) or **nose only** (animals exposed via inhalation ports). Report one of: “inhalation - whole body” or “inhalation - nose only”.

If the route is neither of the three (e.g., intraperitoneal, subcutaneous), then report: “other”.

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+++Hazard Classification+++:

This is only required/available for repeat dose toxicity and some sensitization records (Local Lymph Node Assay only).

First, determine the type of study. Choose one of the following: repeated dose toxicity, murine local lymph node assay (LLNA), or guinea pig sensitisation test (Magnusson Kligman, GPMT, Buehler).

If the study is a repeated dose toxicity study, determine the route of administration used for the in vivo animal test. Choose one of the following primary routes: oral, dermal, or inhalation. Then determine the no adverse effect level (NOAEL) in mg/kg body weight (bw)/day (for oral and dermal studies) or the no adverse effect concentration (NOEC) in mg/L/6 hours (for inhalation studies).

Classify the NOAEL hazard as low, moderate, or high by comparing it to the ranges below. The ranges below are the internal hazard classification criteria used by evaluators in Health Canada for assessing toxicological repeated dose studies. Return the hazard as "low", "moderate" or "high".

- If the route is oral:
 - If the duration is 28 days (4 weeks):
Low hazard: >300 mg/kg bw/day
Moderate hazard: 30-300 mg/kg bw/day
High hazard: <30 mg/kg bw/day
 - If the duration is 90 days (13 weeks):
Low hazard: >100 mg/kg bw/day
Moderate hazard: 10-100 mg/kg bw/day
High hazard: <10 mg/kg bw/day
- If the route is dermal:
 - If the duration is 28 days (4 weeks):
Low hazard: >600 mg/kg bw/day
Moderate hazard: 60-600 mg/kg bw/day
High hazard: <60 mg/kg bw/day
 - If the duration is 90 days (13 weeks):
Low hazard: >200 mg/kg bw/day
Moderate hazard: 20-200 mg/kg bw/day
High hazard: <20 mg/kg bw/day
- If the route is inhalation as a vapour or aerosol:
 - If the duration is 28 days (4 weeks):
Low: >3.0 mg/L/6 hours
Moderate: >0.6-3.0 mg/L/6 hours
High: <0.6 mg/L/6 hours
 - If the duration is 90 days (13 weeks):
Low: >1.0 mg/L/6 hours
Moderate: >0.2-1.0 mg/L/6 hours
High: <0.2 mg/L/6 hours
- If the route is inhalation as a dust:
 - If the duration is 28 days (4 weeks):
Low hazard: >0.6 mg/L/6 hours
Moderate hazard: >0.06-0.6 mg/L/6 hours
High hazard: <0.06 mg/L/6 hours
 - If the duration is 90 days (13 weeks):
Low: >0.2 mg/L/6 hours
Moderate: >0.02-0.2 mg/L/6 hours
High: <0.02 mg/L/6 hours

If the study is a murine local lymph node assay (LLNA, e.g., OECD 429) for skin sensitisation, identify the EC3 value (reported in units of %) from the PDF document and compare it to the classification ranges below. Return the hazard as "low", "moderate", "high", or "very high".

Low: >10%
Moderate: 1.0-10.0%
High: 0.1-1.0%
Very high: <0.1%

If there is no EC3 value reported in the PDF document, then confirm that the document states the stimulation index (SI) did not exceed 3.00 during the study and return "negative". If the SI did exceed 3.00 at one of the test concentrations, but no EC3 value is reported, return "positive".

If the study is a skin sensitisation study in guinea pigs (e.g., guinea pig maximisation test; GPMNT or Buehler test), identify whether the document describes the test item as a "weak", "mild", "moderate", "strong", or "extreme" sensitiser. If so, report the descriptor in field #. If not, leave field # blank.

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+++Critical Effects+++:

For repeated dose records only. Identify the critical treatment-related, adverse effects that led to the selection of the NOAEL (or NOAEC for inhalation).

Critical effects are the specific adverse findings observed at the lowest dose/concentration at which adverse effects occurred (LOAEL/LOAEL(C)) and are typically described in the Discussion, Results, or Conclusion sections of the report.

These effects are the defining reasons the NOAEL was selected. They are the treatment-related adverse effect(s) first seen at the LOAEL, while they were absent or non-adverse at the NOAEL dose/concentration.

Look for critical effects in these types of endpoints and describe them directly as reported:

- Organ weights (e.g., kidney weight, liver weight)
- Organ histopathology / tissue pathology such as degeneration, necrosis, hypertrophy, hyperplasia, inflammation, fibrosis, vacuolation, etc.
- Clinical chemistry parameters (e.g., bilirubin, ALT, AST, creatinine, albumin)
- Hematology parameters (e.g., WBC count, hemoglobin, platelet count, etc.)
- Urinalysis parameters (e.g., protein, specific gravity, blood, etc.)

Examples of reporting outputs:

- “increased absolute and relative kidney weights”
- “decreased bilirubin”
- “centrilobular hepatocyte hypertrophy”

If multiple critical effects are identified, summarize each separately.

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+++Maximum Dose(s) Tested+++:

Identify the maximum (highest) dose or concentration tested in the study. The format and context will vary depending on the type of study (in vivo animal toxicity vs. in vitro genotoxicity). Use the criteria below to extract the correct information and report it in the structured format at the end.

If the test is an in vivo animal toxicity test, identify the dose series reported in the methods or study design section. Look for numeric dose series like “0, 10, 100, 1000 mg/kg bw/day,” “50, 200, 800 mg/kg/day,” or similar. Doses may be described per sex, per day, or per animal group. Determine the highest dose listed in that series and report that highest dose as the maximum dose tested.

Examples you might find:

- 0, 10, 100, 1000 mg/kg bw/day → maximum dose = 1000 mg/kg bw/day
- 20, 100, 500 mg/kg/day → maximum dose = 500 mg/kg/day
- Dose listed only once at “1000 mg/kg/day” if no lower doses specified (i.e., a limit test) → maximum dose = 1000 mg/kg/day

If the test is an in vitro genotoxicity test (e.g., Ames, chromosomal aberration assays), the test item is applied at multiple concentrations (e.g., µg/plate, µg/mL). Different conditions (e.g., with and without metabolic activation like S9, or limited by things like cytotoxicity or formation of precipitates) and strains may have separate dose series. Identify all concentration series reported for each condition in the study. Typical wording: “tested at 50, 158, 500, 1580, 5000 µg/plate”. There may be multiple sets if the test is run with and without S9 or across strains.

For each concentration series, determine the highest concentration tested in that series. Report the range of maximum concentrations across all series.

Examples you might find:

- Condition 1 series: 5, 50, 158, 500, 1580, 5000 µg/plate → max = 5000
- Condition 2 series: 10, 100, 333, 1000, 2500 µg/plate → max = 2500
- Report range: 2500–5000 µg/plate

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+++Test Item (Technical Grade or Analytical) Purity+++:

Identify the analytical purity (expressed as a percentage) of the test item used in the study. In typical toxicology study reports, the test item purity is reported in the Test Item Identity or Materials section and usually appears as a percentage (%) indicating the technical grade purity of the substance supplied to the laboratory. The information may be presented in various ways, including: "Purity: ##.##%", "Test article purity: ###%", "Analytical purity: ##.%", "Assay: ###%", "Technical grade: ###%". Report the purity percentage as the number value and unit percentage.

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+++Test Item Concentration / Dilution+++:

Identify and extract the concentration(s) of the test item used in dermal exposure portions of the study. Look in sections like *Methods*, *Test Procedure*, *Exposure/Dosing*, or *Treatment Conditions* where concentrations are given. Concentrations may appear in tables or narrative text and units may be written as “% w/w” — percent weight/weight, “% w/v” — percent weight/volume, “% v/v” — percent volume/volume.

Do not extract in vivo oral doses where the dose is expressed as mg/kg bw/day, because those are dose levels already adjusted on a per-kg weight basis, not dilution concentrations.

For dermal tests, the concentration refers to the percentage (%) of test item in a vehicle (w/w, w/v, or v/v) used to apply the test item to the skin.

- If the test is a guinea pig skin sensitisation test (i.e., GPMT, Magnusson Kligman or Buehler test, OECD 406), Identify the induction exposure concentration(s) used during the induction phase and identify the challenge (elicitation) exposure concentration(s) used during the challenge phase. Report both concentrations in units of % w/w, w/v, or v/v and report “induction: [induction exposure concentration%] + challenge: [challenge exposure concentration]” in field #.
- If the test is a murine local lymph node assay (i.e., LLNA, OECD 429), multiple concentrations are typically tested in a series. Identify all of the concentrations tested across the dose series applied topically in % units (e.g., 10%, 25%, 50% w/v).
- If the test is a human patch test, find the tested concentration of the substance expressed as a percentage (%) in the vehicle.

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+++Vehicle / Solvent+++:

Identify the vehicle or solvent in which the test item was dissolved, suspended, or otherwise delivered for administration to animals or test subjects. In toxicology reports, this information is typically found in the Test Item Identity section, Materials and Methods, or Study Design sections.

Locate the text that describes the vehicle or solvent used for the test item. Common wording includes phrases such as “administered in”, “dosed in”, “suspended in”, “formulated in”, or “vehicle:”. Extract the specific vehicle or solvent name exactly as stated (e.g., saline, corn oil, PEG 400). If multiple vehicles are mentioned in different parts of the report (e.g., separate formulations for different routes or phases), extract all applicable vehicle names and list them. Ignore any mentions of vehicle controls unless they specify the actual vehicle substance used for the test item.

Examples of common vehicles or solvents include:

Water

Normal saline (0.9% sodium chloride)

Phosphate-buffered saline (PBS)

Buffered solutions (e.g., Tris buffer)

Dimethyl sulfoxide (DMSO)

Polyethylene glycol (e.g., PEG 400, PEG 300)

Propylene glycol

Ethanol

Methyl cellulose (e.g., 0.5% methylcellulose)

Hydroxypropyl methylcellulose

Hydroxyethylcellulose

Hydroxypropyl- β -cyclodextrin (HP- β -CD)

Polysorbate 80 (Tween 80)

Sodium dodecyl sulfate (SDS)

Cremophor® EL / Cremophor formulations

Solutol HS-15

Pluronic® F-127

Kollidon®

Vitamin E TPGS

Corn oil

Olive oil

Sesame oil

Fat emulsions

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+++Sample size (number of test animals)+++:

Sample sizes can only be obtained from in vivo studies. In vitro studies are considered not applicable for retrieving this data.

In in vivo animal studies, test items are administered to treatment groups of male and/or female animals at three or more dose levels in mg/kg bw/day. There is also typically a negative control group (i.e., a group that receives placebo or 0 mg test item/kg bw/day). Typically groups will contain an equal number of male and female animals, commonly 10 animals per dose per group (5 males and 5 females). Optional satellite or recovery groups can also be used concurrently for the control group and the high dose group to assess reversibility or delayed effects during a post-treatment recovery period, using the same number of animals. Alternatively, a limit test design can also be used which involves only 1 treatment group and 1 control group.

Extract and return the following two pieces of information:

1. Total number of animals used in the study (overall count of all animals across all groups, including any satellite/recovery groups, if present).
2. The number and sex of animals in each study group, including:
 - Control group
 - Each treatment/dose group (e.g., low, mid, high)
 - Any recovery/satellite groups, if reported